Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_failure_replay.
- Best TSPLIB ATSP holdout gap: atsp_failure_replay.
- Best synthetic holdout gap: atsp_random_replay.

Run Metadata

- run_name: run_20260513_184910_s
- started_at_local: 2026-05-13 18:49:10
- finished_at_local: 2026-05-13 18:56:16
- duration_hhmm: 00:07
- duration_seconds: 425.854
- seed_offset: 18000
- replicate_label: s
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.190115	0.009004	0.09956	0.0	0.58	0.0
atsp_random_replay	random	score_only	False	0.189168	0.008454	0.098811	0.877558	0.58	0.012769
atsp_failure_replay	failure	score_only	False	0.181339	0.009004	0.095172	0.745308	0.58	0.012437
atsp_failure_replay_compression	failure	novelty_gate	True	0.193143	0.009004	0.101074	0.844675	0.58	0.012371

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.190115, synthetic 0.009004, combined 0.09956.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.190115 across 4 instances; family means: ft=0.319768, ftv=0.33478, p=0.003915, ry=0.101997.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.189168, synthetic 0.008454, combined 0.098811.
- Accepted-epoch count 4, mean accepted code novelty 0.877558, and final complexity 0.58.
- Adaptation efficiency 0.012769 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.189168 across 4 instances; family means: ft=0.319768, ftv=0.33478, p=0.004982, ry=0.097143.
- Panel synthetic_holdout mean gap 0.008454 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.181339, synthetic 0.009004, combined 0.095172.
- Accepted-epoch count 5, mean accepted code novelty 0.745308, and final complexity 0.58.
- Adaptation efficiency 0.012437 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.181339 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004093, ry=0.063237.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1903, "family": "ftv", "name": "ftv38", "optimality_gap": 0.243791}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.193143, synthetic 0.009004, combined 0.101074.
- Accepted-epoch count 4, mean accepted code novelty 0.844675, and final complexity 0.58.

- Adaptation efficiency 0.012371 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.193143 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004093, ry=0.110456.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1903, "family": "ftv", "name": "ftv38", "optimality_gap": 0.243791}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}, {"best_known_cost": 1286, "cost": 1478, "family": "ftv", "name": "ftv33", "optimality_gap": 0.1493}]

Judge Appendix

Comparative Summary of Replay Conditions (ATSP Benchmark)

Condition	Replay Mode	Synthetic Gap	Transfer Gap (TSPLIB + Synth)	TSPLIB Gap	Code Novelty Mean	Complexity	Notes on Transfer & Novelty
atsp_no_replay	none	0.009004	0.09956	0.190115	0.0	0.58	Baseline; moderate transfer, no code novelty
atsp_random_replay	random	**0.008454**	0.098811	0.189168	0.88	0.58	Slightly better synthetic gap; transfer gap similar to no replay but code novelty high
atsp_failure_replay	failure	0.009004	**0.095172**	**0.181339**	0.75	0.58	Best TSPLIB gap and best transfer gap; moderate code novelty
atsp_failure_replay_compression	failure + compression	0.009004	0.101074	0.193143	0.84	0.58	Slightly worse transfer (TSPLIB gap up); high code novelty

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Key Interpretations

- **Transfer Performance**
- The **atsp_failure_replay** condition yields the best transfer outcomes on held-out TSPLIB instances (gap 0.181339) and also the best combined transfer gap (TSPLIB + synthetic), outperforming no replay and random replay despite somewhat lower code novelty.
- **Random replay** achieves the best synthetic holdout gap (0.008454) but transfer to TSPLIB is marginally worse than failure replay.
- Compression-aware failure replay shows a degradation in transfer gap compared to failure replay without compression, suggesting compression pressure may impair transfer despite increased code novelty.

- **Code Novelty vs Transfer**
- **Random replay and compression-aware failure replay** both exhibit high code novelty (~0.84–0.88 mean), yet their transfer gaps are not better than failure replay with moderate novelty (~0.75).
- This indicates that increased lexical/code novelty does not necessarily translate to better transfer or algorithmic advancement under the given deterministic metrics.

- **Complexity and Behavior Profile**
- All conditions maintain the same mean complexity (0.58) and balanced behavior profile, isolating replay mode and code novelty as the main varying factors.

- **Replay Mode Effects**
- No replay (baseline) performs worse on transfer than failure replay, showing replay helps adaptation.
- Failure replay outperforms random replay on key transfer gaps, despite random replay showing higher novelty.
- Compression pressure added to failure replay degrades transfer gap markedly.

- **Optimality Gaps (Held-out TSPLIB families)**
- Failure replay reduces mean heldout TSPLIB family gap the most, particularly improving on the 'ry' family (0.063 vs. ~0.10 with others).
- Worst-case instance gaps remain high across all conditions for certain families (ftv, ft), indicating room for improvement beyond replay modes tested.

Conservative Conclusions

- **Failure replay condition yields the most effective transfer to real-world ATSP benchmarks (TSPLIB), achieving the lowest optimality gaps and best transfer gaps.**
- **Random replay produces higher code novelty but does not improve important deterministic transfer metrics, indicating lexical novelty is not evidence of algorithmic invention here.**
- **Compression-aware failure replay results in elevated transfer gaps despite high code novelty, suggesting compression pressure may reduce transfer effectiveness.**
- **No-replay baseline performs worst on transfer metrics, supporting the utility of replay mechanisms focused on failure cases.**

Recommendations for Benchmark Evaluation

- Prioritize **failure replay** for future algorithmic tuning and transfer studies due to superior transfer performance.
- Do not conflate observed high code novelty in random or compression replay modes with actual algorithmic progress unless accompanied by improved transfer or optimality results.
- Further investigate why compression pressure impairs transfer and whether alternative archive management strategies might maintain transfer benefits with compression.